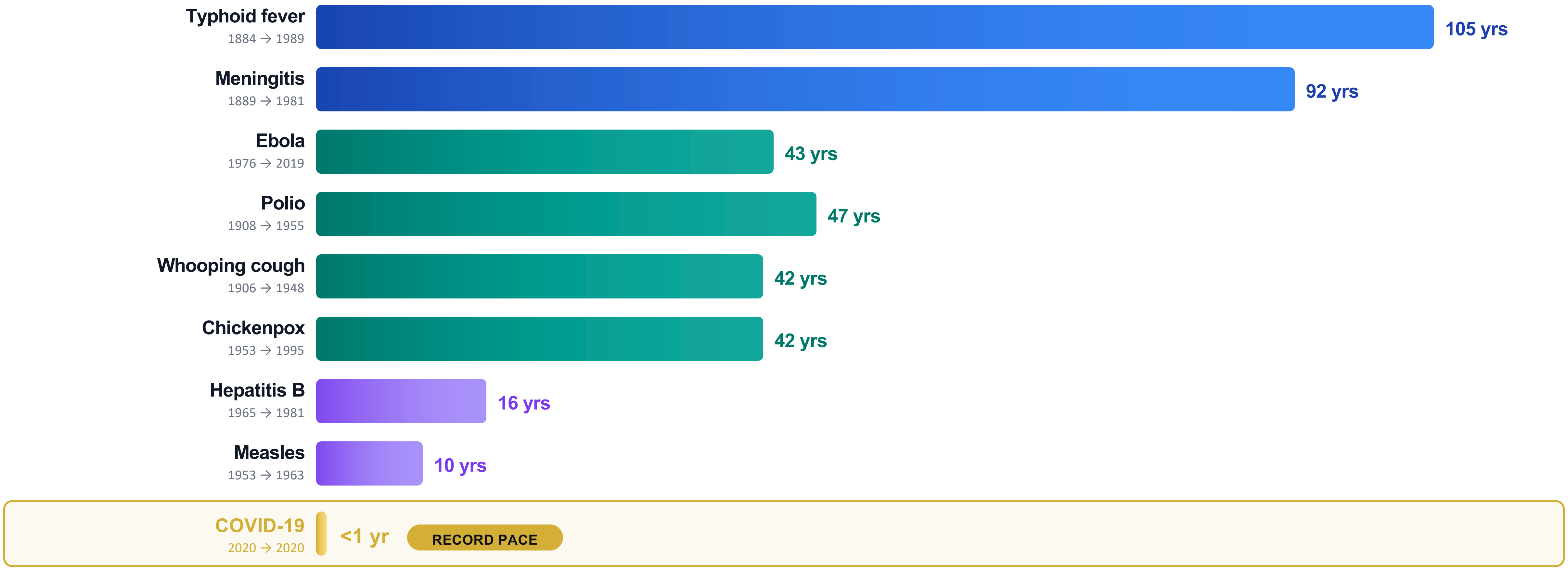
The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

A Data-Driven Overview of How Vaccines Transformed Public Health (1796–2020)

SOURCE: OUR WORLD IN DATA

COVID-19 vaccines were developed in under 1 year — compared to 105 years for typhoid



Modern tools have collapsed development timelines

mRNA technology · Genome sequencing · Virus-like particles · Advanced microscopy

Vaccines eliminated diphtheria, smallpox, and polio; others show dramatic but incomplete reductions

Disease	Pre-Vax Cases	Post-Vax Cases	Case Red.	Pre-Vax Deaths	Death Red.	per million/yr	Status
Diphtheria	158	0	100%	13.7	100%		✓
Smallpox	250	0	100%	2.9	100%		✓
Acute polio	141	0	100%	10	100%		✓
Measles	3,044	0.2	99.99%	2.5	100%		✓
Rubella	242	0.04	99.98%	0.09	100%		✓
Pertussis	1,534	52	96.6%	30.8	99.7%		Near
Hepatitis A	465	51	89%	0.5	88.7%		⚠
Acute hepatitis B	273	44	83.9%	1	83.6%		⚠
Pneumococcal	233	139	40.5%	24	31.3%		⚠

Source: Roush & Murphy (2007); Our World in Data

From 100–1,000+ cases per 100,000 before 1963 to near zero after 1980 mandate

	Pre-1940	1940s	1950s	1960s	1970s	1980–2000	2000–2022
Northeas ME, NH, VT,	500+	400	350	120	30	<1	0
Mid-Atla NY, NJ, PA	800+	700	600	150	20	<1	0
Southeas FL, GA, SC, N	1000+	900	700	200	50	<1	0
Midwest IL, OH, MI, IN	700+	600	500	100	25	<1	0
South Ce TX, LA, AR, O	900+	800	650	180	40	<1	0
Mountain CO, AZ, NM,	500+	450	400	90	15	<1	0
Pacific CA, WA, OR	600+	550	450	130	20	<1	0

Cases per 100,000

0 100 300 700 1,000+

Sporadic outbreaks still occur,
but endemic transmission was eliminated

1963: Vaccine

1971: MMR

1980: Mandate

Source: Our World in Data; US CDC Measles Data (1929–2022)

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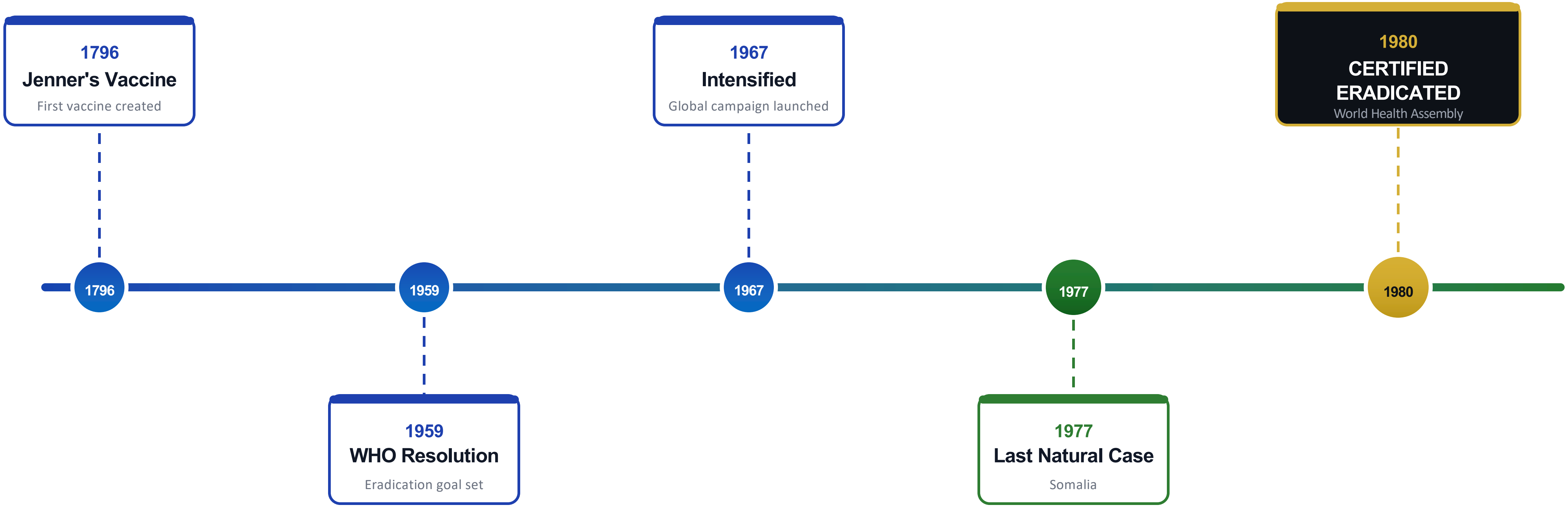
A global vaccination campaign achieved what no other has — complete eradication

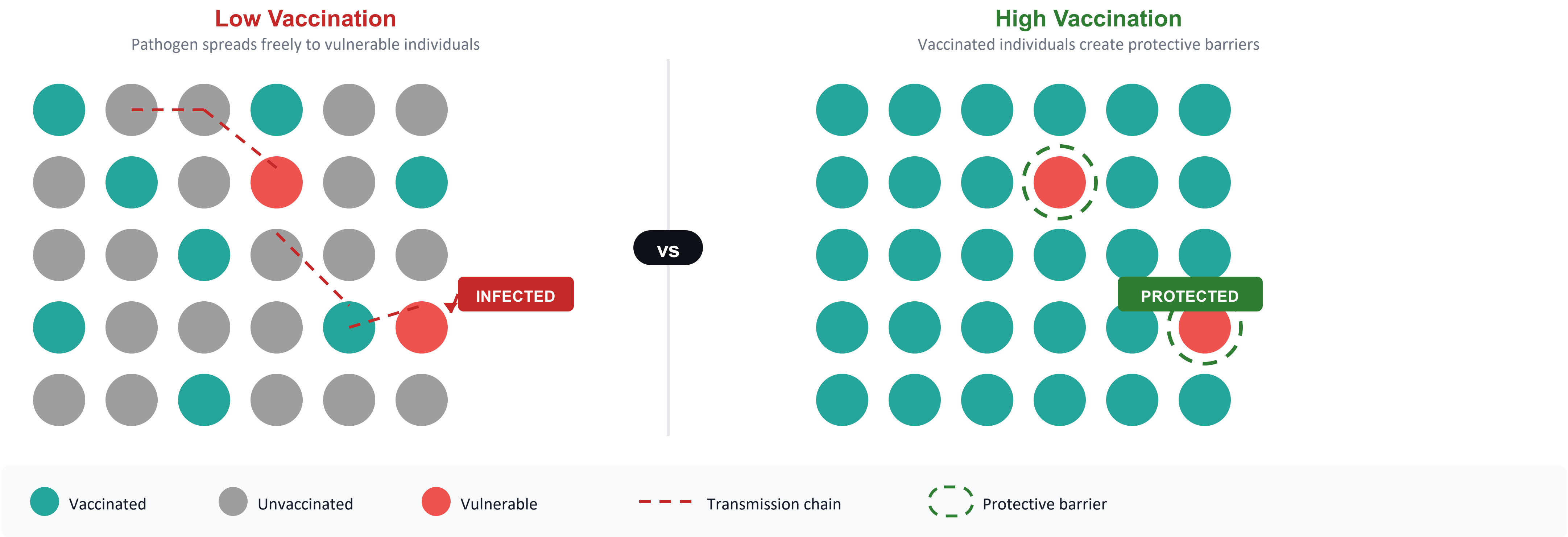


Why smallpox was uniquely suited for eradication

Infected only humans (no animal reservoir) · Symptoms were easily distinguishable · Vaccine was cheap and effective · No carrier state

The only human disease ever fully eradicated through vaccination

When enough people are vaccinated, pathogens cannot spread — shielding those who cannot be vaccinated



Who herd immunity protects:

Newborns too young to vaccinate · Immunocompromised individuals · Cancer patients on chemotherapy · Those with vaccine allergies

Herd immunity threshold varies by disease: Measles ~95% · Polio ~80% · Influenza ~40-60%

Significant disease burden remains for several vaccine-preventable diseases

Pneumococcal Disease

Case Reduction

40.5%

40.5%

Death Reduction

31.3%

Pre-vax: 233 cases/million → Post-vax: 139 cases/million

Hepatitis A

Case Reduction

89%

89%

Death Reduction

88.7%

Pre-vax: 465 cases/million → Post-vax: 51 cases/million

Acute Hepatitis B

Case Reduction

83.9%

83.9%

Death Reduction

83.6%

Pre-vax: 273 cases/million → Post-vax: 44 cases/million

Key Barriers to Complete Vaccine Coverage

Access Gaps

Globally, not everyone has access to vaccines, especially in low-income regions

Supply Shortages

Vaccine supply shortages put children at risk in developing countries

Trust Challenges

Poor scientific understanding can reduce trust and uptake of vaccination programs

Vaccine Hesitancy

Misinformation and complacency erode confidence even where vaccines are available

Incomplete coverage leaves vulnerable populations at risk — even one unvaccinated individual can trigger an outbreak

Key Takeaways

Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

100% elimination achieved
Diphtheria, smallpox, and polio fully eliminated in the US through vaccination

Development timelines collapsed
From over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA and genomic tools

Measles collapsed across all 50 states
Near-zero cases after 1980 kindergarten vaccination mandate; only sporadic outbreaks since

Smallpox: the only eradicated human disease
Certified eradicated in 1980 after a global campaign — proof that vaccination can end diseases

Gaps persist
Pneumococcal (40.5%), hepatitis A (89%), hepatitis B (83.9%) — coverage gaps remain

CALL TO ACTION

Improve Access
Ensure equitable vaccine distribution globally

Strengthen Coverage
Close remaining gaps in vaccination rates

Build Trust
Combat misinformation and hesitancy